

Canada’s Adhesive Dependency

Net Import Reliance, Certification Lock-In, and the Adhesive Chemistry of Canada’s Forest Products Value Chain

AUTHOR

John Crandlemire, P.Eng.

Independent research

DATE

June 2026

High River, Alberta

FORMAT

Independent analysis

KEYWORDS

- forest products
- adhesive resins
- supply chain resilience
- engineered wood
- mass timber
- formaldehyde resins
- Prince Rupert
- value-added manufacturing

Executive summary

Canada produces approximately 9.9 billion ft² of structural panels annually yet imports \$1.76 billion CAD in adhesive resins and prepared adhesives (HS 3506+3909), with 72% (\$1.28 billion) from the United States, a structural vulnerability in a sector contributing \$27 billion to GDP. Domestic converter capacity (Hexion Edmonton, St. Romuald) covers part of national panel output; certification lock-in (CANFER, CSA O112, CCMC) reinforces US supply even when Canadian alternatives are available. We combine Statistics Canada CIMA trade data (2014–2024) with a bottom-up resin demand model covering OSB, plywood, particleboard, and fibreboard (\$1.16 billion at mid-range prices), which brackets the forest-sector adhesive import exposure between \$1.2 billion and \$1.8 billion annually. Phenolic (HS 390940, 92% US), polyurethane (HS 390950, 80% US), and phenol feedstock (HS 290711, 99% US) disaggregation reveals mass-timber and glulam exposure distinct from commodity lumber trade. Pacific Northwest routing, 2025–2026 tariff shocks, and CANFER product-type lock-in are analyzed with Prince Rupert as an illustrative converter-scale site. Policy levers include IFIT eligibility, supplier-substitution protocols, and bio-based pathways.

1. Introduction

Canada is the fourth-largest forest products exporter in the world (Genome Québec 2019) and maintains the largest forest product trade balance globally (Forisk Consulting 2024). It produces approximately 9.9 billion square feet of structural panels annually (OSB and softwood plywood combined, 2024 estimate annualized from APA data) (Fastmarkets 2024a), supports a rapidly scaling mass timber sector (Forest Products Association of Canada 2024a), and hosts a wood panelling manufacturing industry generating \$11.8 billion in revenue in 2025 (IBISWorld 2025). Yet for all this output, Canada remains structurally dependent on imported adhesive resins: domestic panel-grade capacity is limited to Hexion's Edmonton and St. Romuald plants (plus niche structural and bio-binder producers), while imports of prepared adhesives and bulk resins exceed \$1.7 billion CAD annually.

The shortfall in domestic industrial-scale capacity in forest adhesive chemistry, relative to national panel and mass-timber output, represents a structural vulnerability in an industry that employs more than 375,000 people (Natural Resources Canada 2024a) and contributes \$27 billion annually to GDP (Natural Resources Canada 2024b). The dependency takes four measurable forms: net import reliance, single-supplier concentration, Pacific Northwest routing for western mills, and certification lock-in; Canada retains converter-scale synthesis at two Hexion plants. As long as most of the glue that makes Canadian forest products “value added” is manufactured in the United States (or imported as bulk chemistry from Asia and formulated abroad), Canada captures less of the formulation margin in its wood-composites chain.

Our contribution is fourfold: (i) a product-level quantification of resin chemistry and loading rates across Canada's major engineered wood categories; (ii) a supply-chain analysis of North American resin concentration and Canada's feedstock-conversion gap; (iii) an analysis of Canadian certification and CANFER/CSA approval pathways that constrain drop-in resin substitution and new domestic supply; and (iv) a site-specific industrial policy case for Prince Rupert as a domestic resin manufacturing node, with explicit data gaps flagged for trade-flow verification.

2. Methods and Data Sources

2.1 Analytical Approach

We combined publicly available industry, government, and technical sources on forest adhesive chemistry, panel manufacturing, and trade policy. Resin loading estimates are drawn from CORRIM life-cycle inventories (CORRIM 2018a; 2018b), manufacturer technical data sheets (Arclin Inc. 2019), and peer-reviewed wood-composites literature (FPInnovations 2020a; Li 2010). Supply-chain claims regarding US producer concentration are based on company disclosures, trade press, and industry association reports (Hexion Inc. 2020; Composite Panel Association 2024; 2025). Site suitability analysis for Prince Rupert draws on municipal industrial development materials (City of Prince Rupert 2024; Business in Vancouver 2024).

2.2 Trade Data and Bottom-Up Model

Import values for HS 3506 (prepared adhesives) and HS 3909 (amino/phenolic resins and polyurethanes in primary forms) are drawn from Statistics Canada's Canadian International Merchandise Trade (CIMT) bulk HS6 import files, aggregated to HS4 (Statistics Canada 2024). Disaggregated subcodes HS 390940 (phenolic resins), HS 390950 (polyurethanes), HS 290711 (phenol), and HS 290721 (resorcinol) were retrieved from the same bulk files. Values are reported in Canadian dollars on a customs basis. Annual series 2014–2024 were compiled from CIMT HS6 bulk import records and cross-checked against UN Comtrade bilateral totals where available. These codes are broader than forest-panel resins alone; HS 3909 in particular includes polyurethanes used outside wood composites. We report combined totals with explicit scope notes and do not treat HS 3506 and HS 3909 as mutually exclusive where customs classification may overlap.

Structural panel production volumes use APA data reported via Fastmarkets for the first half of 2024, annualized (doubled) as a proxy for full-year output (Fastmarkets 2024a). This introduces seasonality uncertainty; full-year APA yearbook data are preferred when available. Particleboard and fibreboard volumes are drawn from the FAO Forestry production database (Canada, 2022, the latest complete year): 1.62 million m³ of particle board (excluding OSB) and 1.28 million m³ of fibreboard, which in Canada is dominated by MDF (Food and Agriculture Organization of the United Nations 2024). A bottom-up resin demand model, reproduced in the supplementary workbook, applies CORRIM loading rates and mid-range resin prices to all four product categories.

2.3 The Bottom-Up Demand Model

The model is three equations. For each panel product p with annual production volume V_p (m³), panel density ρ_p (kg/m³), and dry-resin loading fraction λ_p :

$$M_p = V_p \cdot \rho_p \cdot \lambda_p \quad (1)$$

gives the embedded dry resin mass. Input parameters (Table 1) draw loading fractions and densities from CORRIM life-cycle inventories (CORRIM 2018b; 2018a), structural panel volumes from APA 2024 data (Fastmarkets 2024a), and particleboard and fibreboard volumes from FAO Forestry statistics (Food and Agriculture Organization of the United Nations 2024):

Table 1: Bottom-up model inputs and computed dry resin mass by product, per (1).

Product	V_p (m ³ /yr)	ρ_p	λ_p	P_p (\$/kg)	M_p (t)
OSB (PF/pMDI)	7,136,000	650	0.040	3.80	185,500
Softwood plywood (PF)	1,580,000	480	0.030	3.80	22,800
Particleboard (UF)	1,625,000	746	0.060	2.26	72,700
Fibreboard / MDF (UF)	1,277,000	750	0.095	2.26	91,000
Total					372,000

The annual resin cost applies delivered prices P_p by resin family (PF/pMDI \$3.80/kg, range \$3.30–4.30; UF \$2.26/kg, range \$2.00–2.60 (EcoSynthetix Inc. 2023a; CORRIM 2018b)):

$$K = \sum_p M_p \cdot P_p = \$792\text{M (structural panels)} + \$370\text{M (PB/MDF)} \approx \$1.16\text{B/yr.} \quad (2)$$

The price ranges move K between \$1.02B and \$1.32B. The import-dependency ratio for year t is measured directly from customs data:

$$U_t = \frac{I_t^{\text{US}}}{I_t}, \quad U_{2023} = \frac{\$1.28\text{B}}{\$1.76\text{B}} = 72\%. \quad (3)$$

Because (2) excludes glulam, CLT, LVL, and I-joist adhesives while customs totals I_t include non-forest uses of the same HS codes, the two estimates bracket the forest sector's annual adhesive cost D : $K \leq D \leq I_t$, i.e. \$1.2B–\$1.8B annually (Figure 3).

2.4 Limitations

This is a policy and supply-chain analysis; financial feasibility, including a discounted-cash-flow model for a Prince Rupert resin plant, sits outside its scope. HS 3506 and HS 3909 include non-forest-adhesive uses, so trade totals may overstate forest-specific imports; the bottom-up model excludes glulam, CLT, LVL, and I-joist adhesives and therefore understates total sector resin demand. Structural panel production is annualized from H1 2024 APA data and may miss full-year seasonality; particleboard and fibreboard volumes are 2022 FAO Forestry values, the latest complete year, and the fibreboard aggregate includes a hardboard share for which the MDF loading rate overstates resin demand somewhat. Company acquisition timelines (Kronospan/Hexion La Grande, December 2025) reflect trade-press reporting. Resin price scenarios use published mid-range estimates (EcoSynthetix Inc. 2023a; CORRIM 2018b) rather than mill-gate contract prices. The Prince Rupert site case is geographic screening only; permitting timelines, offtake contracts, and capital cost are outside the model.

3. Quantitative Results

3.1 Import Dependence

Canada's combined imports of HS 3506 and HS 3909 totalled approximately \$1.76 billion CAD in 2023 (Statistics Canada 2024). The United States supplied \$1.28 billion of this total, 72% of combined import value per (3). Disaggregated by code: HS 3506 (prepared adhesives) imports were \$679 million CAD, with 81% from the US; HS 3909 (bulk resins) imports were \$1.08 billion CAD, with 67% from the US. Combined import value rose from \$1.06 billion in 2014 to \$1.82 billion in 2024; the US share of combined value declined from 86% to 71% over the same period as non-US suppliers, especially within HS 3909 polyurethanes, gained share.

The decade-long trend in Figure 1 shows import value growing 72% while US dominance persists: prepared adhesives (HS 3506) remain 81–86% US-sourced throughout, whereas bulk resins (HS 3909) show greater supplier diversification after 2021. For a forest-sector executive, the practical implication is dual exposure: western panel mills depend on Oregon and Washington formulation plants for HS 3506 supply, while mass-timber pMDI systems (HS 390950) face a widening Asian supplier base that leaves certification lock-in intact (Figure 2).

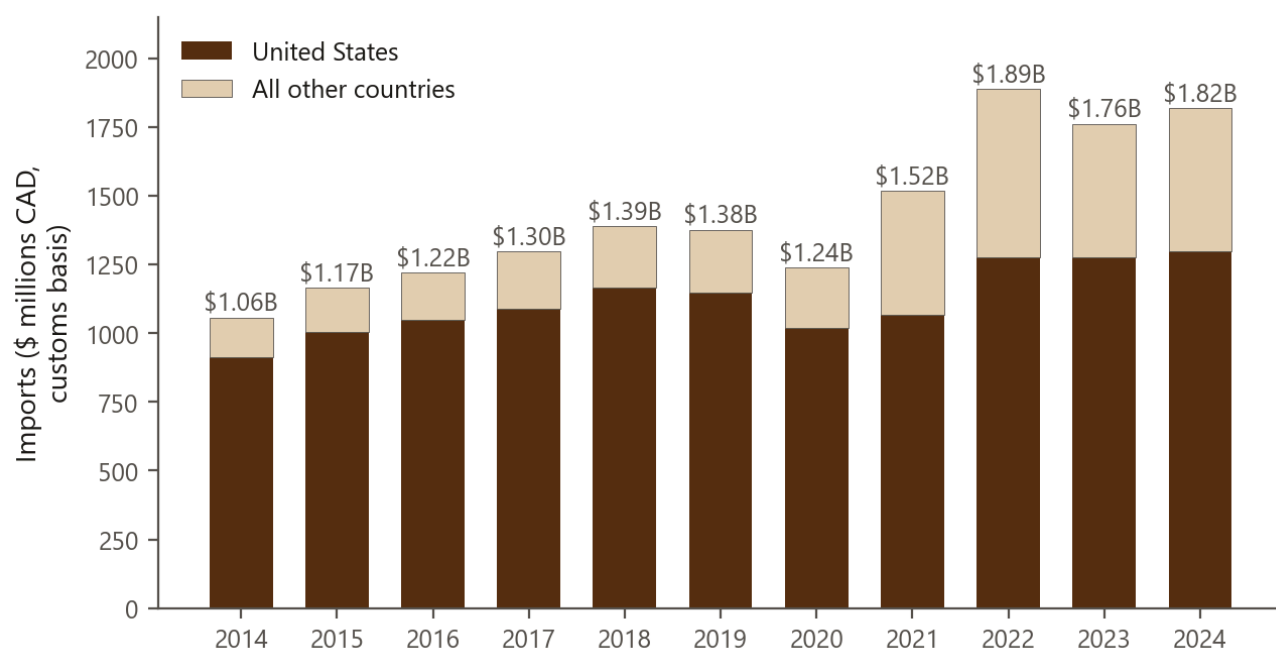


Figure 1: Canadian imports of prepared adhesives (HS 3506) and bulk resins (HS 3909), 2014–2024, split by origin. Total import value grew 72% over the decade while the US remained the dominant supplier; year totals are labelled in \$ billions. Source: Statistics Canada CIMENT HS6 bulk import files.

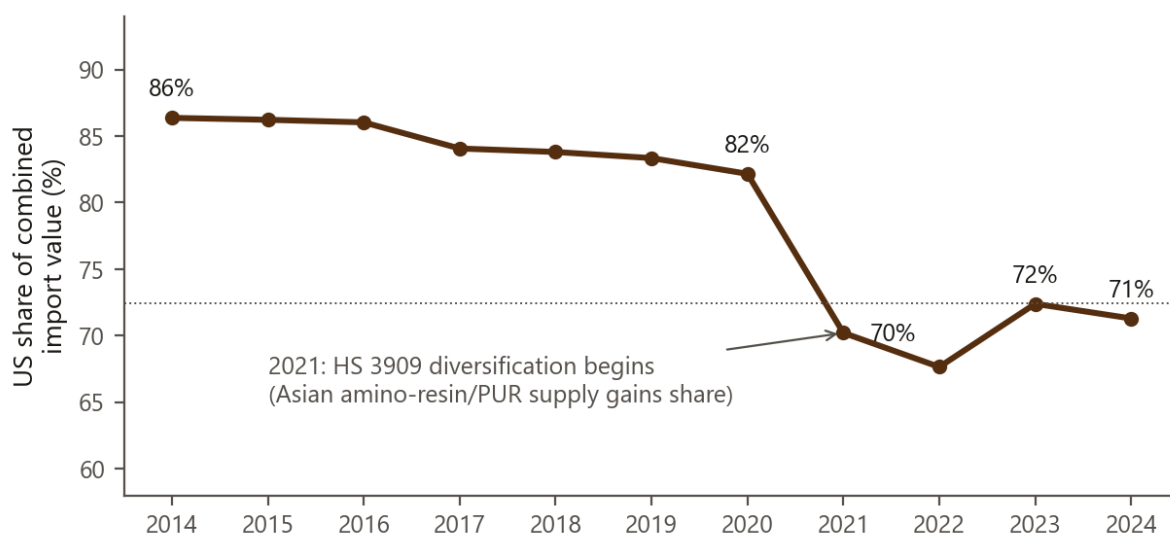


Figure 2: US share of combined HS 3506 + 3909 import value, 2014–2024, applying (3) per year. The step decline in 2021–2022 is driven by Asian amino-resin and polyurethane supply within HS 3909; the HS 3506 prepared-adhesives share has remained at 81–86% US throughout. Source: Statistics Canada CIMENT.

Table 2: Canada adhesive and resin imports by HS code, 2023 (CAD, Statistics Canada CIMENT customs basis).

HS Code	Total Imports	US Imports	US Share
3506, Prepared adhesives	\$679M	\$551M	81%
3909, Bulk resins/PUR	\$1.08B	\$725M	67%
Combined	\$1.76B	\$1.28B	72%

These figures confirm the paper's central claim: Canada imports well over \$1 billion annually in adhesive chemistry, with the United States as the dominant supplier. The magnitude substantially exceeds prior qualitative estimates of “hundreds of millions.”

3.2 Bottom-Up Resin Demand (Four Product Categories)

Applying (1) and (2) to the volumes in Table 1 yields an estimated 185,500 tonnes of dry resin for OSB, 22,800 tonnes for plywood, 72,700 tonnes for particleboard, and 91,000 tonnes for fibreboard, about 372,000 tonnes in total. At the resin-family prices in Methods, the mid-range annual cost is approximately \$1.16 billion CAD: \$792 million for structural panels at \$3.80/kg and \$370 million for particleboard and fibreboard at \$2.26/kg. Glulam, CLT, LVL, and I-joist adhesives sit outside the model and would add further demand.

The four-category estimate (\$1.16B) represents roughly 66% of combined HS import value (\$1.76B). The remaining gap has identifiable components: polyurethane and prepared-adhesive imports serving non-forest end uses, mass-timber adhesives outside the model, and domestic Hexion production that substitutes for imports on some volumes while its phenol feedstock still arrives from the US. Figure 3 shows how the independent methods bracket the forest sector's adhesive cost: (2) is a lower bound (four panel categories); customs totals I_t are an upper bound (all end uses of HS 3506/3909). The resin-family price ranges move K between \$1.02B and \$1.32B without changing the bracket logic.

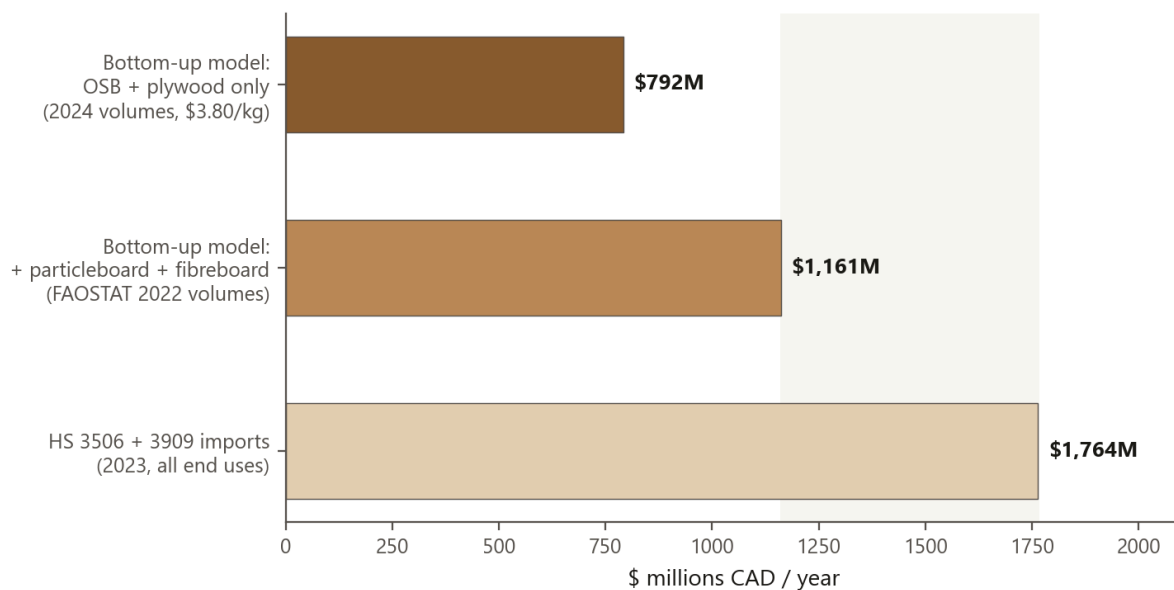


Figure 3: Bracketing the forest sector's annual adhesive cost. The bottom-up model of (2) is a lower bound (structural panels alone, then all four panel categories); the customs total is an upper bound (all end uses of HS 3506/3909). The forest-sector figure lies in the shaded interval.

3.3 Foreign Content by Weight

Resin loading rates translate import dependence into a physical fact about finished products (Table 4). For OSB, CORRIM-based models apply approximately 4% dry resin solids by panel weight; softwood plywood averages roughly 3% (CORRIM 2018b). At 2024 Canadian production volumes, OSB and plywood together embed about 208,000 tonnes of dry resin in roughly 5.4 million tonnes of panel product, a blended share near 4% by weight

for structural panels. Adding particleboard and fibreboard raises the embedded total to about 372,000 tonnes per Table 1.

Roughly one kilogram in every twenty-five in a Canadian OSB panel is adhesive chemistry, and particleboard and MDF loadings run 7–12% while commodity lumber carries none. The “Made in Canada” label on engineered wood therefore describes Canadian fibre, labour, and pressing, but a measurable fraction of exported panel mass, and a larger fraction of formulation margin, originates in US resin supply chains, especially for western mills served from Oregon and Washington.

3.4 Comparison to Softwood Lumber Trade

Public discourse on Canada–US forest trade focuses overwhelmingly on softwood lumber: duties, stumpage, Section 232 investigations, and export volumes that routinely exceed \$5 billion USD annually to the United States (Natural Resources Canada 2024c; Miller Thomson 2025). Lumber leaves the mill without structural adhesive; the trade fight is about fibre pricing.

Adhesive imports operate on a parallel track invisible in lumber statistics. Combined HS 3506+3909 imports (\$1.76B CAD, 72% US) supply the chemistry behind panel, particleboard, MDF, and mass-timber manufacturing, the products marketed as Canada’s value-added forest future. Canada exports billions in lumber while importing billions in the glue that binds engineered wood. The two flows are linked by housing cycles and mill economics but separated in customs reporting and media coverage: HS 44 (wood) dominates headlines; HS 3506/3909 (adhesives) pass largely unreported.



Figure 4: Parallel Canada–US forest trade tracks: southbound lumber exports (HS 44, no structural adhesive) versus northbound adhesive and bulk-resin imports (HS 3506/3909) that support Canadian panel and mass-timber manufacturing. Customs reporting and media coverage rarely treat the two flows together.

Policy implication: tariff and trade-diversification strategies aimed at lumber alone leave untouched the south-to-north resin routing that western panel mills depend on and the CANFER certification lock-in documented below.

3.5 Disaggregated Resin and Feedstock Imports (2023)

CIMT HS6 disaggregation clarifies which resin families drive import dependence (Table 3). Phenolic resins in primary forms (HS 390940), the customs line closest to PF and PRF chemistry, totalled \$129 million CAD in 2023, with 92% from the United States. Polyurethanes in primary forms (HS 390950), a proxy for pMDI systems used in CLT and other mass-timber applications, totalled \$314 million CAD, 80% US-sourced. Amino-resin imports within HS 3909 (UF/MUF, excluding phenolic and PUR subcodes) were \$641 million CAD, with a lower US share (55%) as Chinese amino-resin suppliers gained share. Bulk phenol imports (HS 290711) were \$121 million CAD, 99% from the US, consistent with Edmonton Hexion importing phenol while manufacturing PF locally. Bulk resorcinol (HS 290721) was only \$758,000 CAD in 2023; PRF demand is embedded in phenolic and prepared-adhesive codes rather than this feedstock line.

Table 3: Disaggregated adhesive and feedstock imports, 2023 (CAD, CIMT customs basis).

HS Code	Description	Imports	US Share
390940	Phenolic resins (PF/PRF)	\$129M	92%
390950	Polyurethanes (pMDI proxy)	\$314M	80%
3909†	Amino resins (UF/MUF)	\$641M	55%
290711	Phenol feedstock	\$121M	99%
290721	Resorcinol feedstock	\$0.8M	17%

† HS 3909 subtotal excluding 390940 and 390950; full HS 3909 including subcodes equals \$1.08B (67% US).

3.6 Phenol: Canada's Lost Synthesis Step

Bulk phenol (HS 290711) deserves separate treatment because it exposes the feedstock layer beneath PF resin dependence. Canada imported approximately \$121 million CAD of phenol in 2023, 99% from the United States, roughly 60,000 tonnes at ≈\$2/kg CAD landed; the last domestic phenol synthesis unit closed in 1992 (Health Canada / Environment Canada 1997).

Canada hosted the first commercial Hock (cumene) phenol plant in the world in 1952 at Montreal East (Shawinigan Chemicals and British American Oil) (INIS/IAEA 1990), and later operated a second route at Dow's Ladner, BC facility using toluene oxidation to supply BC plywood adhesive demand (Chemcess 2024). By 1976, Gulf Oil Canada at Montreal East (≈27,000 t/y) and Dow Ladner (≈23,000 t/y) supplied phenol to Reichhold, Borden, Hooker, and other PF resin plants across the country. Both units closed by 1992; phenol has been imported ever since. In the mid-1990s, Canada imported 76,000–95,000 tonnes annually, with roughly 85% consumed in phenolic resin manufacture (Health Canada / Environment Canada 1997).

The United States, by contrast, consolidated phenol on the Gulf Coast and in the Midwest as the cumene route became dominant after 1960. INEOS Phenol operates world-scale units at Mobile, Alabama (expanding to 850,000 t/y) and a 907,000 t/y cumene plant at Pasadena, Texas; AdvanSix produces more than one billion pounds per year at Frankford, Pennsylvania (INEOS Phenol 2018; AdvanSix Inc. 2024). US phenol capacity exceeds 2 million tonnes annually, orders of magnitude above Canadian import demand.

Plant economics explain why phenol synthesis has stayed south of the border. World-scale Hock plants integrate cumene, phenol, and acetone at 400,000+ tonnes per year of phenol capacity, with capital intensity typically in the \$500 million–\$1 billion+ range for greenfield US Gulf Coast designs (S&P Process Economics Program design bases; proprietary detail) (S&P Global / IHS Markit 2020). Operating economics depend on benzene and propylene feedstock costs, energy, and, critically, acetone co-product markets (≈0.62 tonnes acetone per tonne phenol). Canada's current phenol import demand (≈60–120 kt/y) is far below world-scale; a new plant would require export outlets or integrated BPA/polycarbonate demand beyond what forest panels alone could support.

For forest adhesive policy, the relevant comparison is therefore resin conversion (formaldehyde + purchased phenol, moderate capital) versus phenol synthesis (integrated petrochemical scale). Hexion Edmonton exemplifies the converter model: local formaldehyde, imported phenol, exported PF/UF. Rebuilding phenol synthesis would be a petrochemical investment decision; restoring domestic PF resilience runs through the converter layer.

3.7 Resorcinol and Specialty PRF Supply

Bulk resorcinol (HS 290721) is a minor customs line for Canada: \$758,000 CAD imported in 2023. PRF demand for glulam and specialty structural bonds is embedded in HS 390940 phenolic resins (\$129M, 92% US) and prepared adhesives (HS 3506). North American resorcinol synthesis ceased when Indspec closed its Petrolia, Pennsylvania plant in July 2017; prior to that, Canada imported resorcinol overwhelmingly from the US. Post-2017, direct imports shifted toward Japan, China, and India, with the US serving as a formulation and re-export hub rather than a producer. Hexion Edmonton manufactures UF/PF without resorcinol synthesis; domestic resorcinol capacity would face a ~50–55 kt/yr global market and is economically irrational at Canadian scale.

3.8 Canadian Panel Production

APA-reported Canadian structural panel output in the first half of 2024 was 4.03 billion ft² OSB and 893 million ft² softwood plywood (Fastmarkets 2024a). Annualized, this implies approximately 9.9 billion ft² of structural panel production, consistent with industry characterizations of Canada as a ~10 billion ft² panel producer.

4. The Adhesive Content of Wood Panels

When a Canadian OSB mill presses strands of aspen or poplar into a structural panel, the wood itself represents only part of what gets sold. Every sheet of OSB, every particleboard cabinet substrate, every MDF moulding, every glulam beam, and every CLT floor panel is, by weight, substantially adhesive. Table 4 summarizes primary resin systems and typical dry-resin loading rates by product category.

Table 4: Primary resin systems and typical loading rates in Canadian wood panel manufacturing.

Product	Primary Resin	Loading	Notes
Particleboard	Urea-formaldehyde (UF)	~9.2%	Liquid, 65% solids (CORRIM 2018a)
MDF	UF or pMDI	7–12%	Core/face variable (Borregaard 2024)
OSB	PF or pMDI	2–6%	Applied to strands (PatSnap Eureka 2024; FPInnovations 2020a)
Glulam / LVL	PRF or PUR	per bond line	Finger joints + laminating (CORRIM 2018b)
CLT	PUR, EPI, or MF	per bond line	Face bond lines (Mass Timber Institute, University of Toronto 2024)

For a standard particleboard panel at 746 kg/m³, a 9.2% loading translates to roughly 68 kg of liquid resin per cubic metre, or approximately 44 kg on a dry solids basis (Covestro 2020). At Canadian OSB and structural panel production rates of approximately 4–5 billion square feet per half-year (Fastmarkets 2024b), and accounting for PF and pMDI loadings in OSB (1.5–6% by weight) (FPInnovations 2020a), aggregate resin demand flowing into Canadian mills annually is enormous, measured in hundreds of thousands of tonnes.

Adhesive is a primary manufacturing input at these volumes, and the overwhelming majority of it arrives from one direction: south, across the border from the United States.

4.1 The Chemistry Behind the Number

All dominant formaldehyde-based resins, UF, MUF, PF, and PRF, share a common chemistry pathway that begins with methanol (CORRIM 2018b). Methanol is reacted over a metal catalyst to produce formaldehyde, which is then condensed with urea, melamine, phenol, or resorcinol to produce the desired thermoset resin. Formaldehyde resins represent the single largest downstream use of formaldehyde globally, with demand tracking building and remodelling markets (Chemical Market Analytics 2024). Precursor chemicals, methanol, urea, melamine, phenol, and resorcinol, are ultimately derived from natural gas and/or crude oil (CORRIM 2018b). Canada produces natural gas in abundance. Yet the refining, synthesis, and formulation of those inputs into panel-grade resins is almost entirely done elsewhere.

5. The North American Resin Supply Chain

5.1 Who Makes the Glue

The North American forest products adhesive market is dominated by a small number of large specialty chemical producers, most headquartered in the United States and operating production facilities in the Pacific Northwest and the Gulf Coast, geographically positioned to serve major panel-producing regions.

The two most significant historical players supplying PF and UF resins to Canadian wood panel producers have been:

Hexion Inc. (Columbus, Ohio). Hexion's Forest Products Division operated approximately 40 facilities globally, with about 24 in North America (Hexion Inc. 2020). Hexion has been characterized as a "primary partner with all of the panel and engineered wood production segments of the industry" (Hexion Inc. 2020). The company's La Grande, Oregon facility manufactured urea-formaldehyde resins specifically for board production (Chemical Week 2025). In December 2025, Hexion sold the La Grande plant to Kronospan, a global wood products manufacturer (Composite Panel Association 2025; Kronospan 2025; EUWID Wood Products & Panels 2025), a transaction that vertically integrates resin supply into a major panel producer's captive supply chain, potentially reducing third-party supply availability to Canadian mills.

Arclin (formerly Georgia-Pacific Resins). Arclin is a materials science manufacturer headquartered in Alpharetta, Georgia, with manufacturing facilities in Portland, OR; Tacoma, WA; and several eastern US locations (Composite Panel Association 2024). Arclin supplies phenol-formaldehyde resins commercially demonstrated for OSB production to Canadian mills including Tolko (Arclin Inc. 2019). In early 2026, Arclin expanded through acquisitions including The Willamette Valley Company and DuPont's Aramids business (Arclin 2026a; 2026b).

The concentration of panel-grade resin manufacturing in the Pacific Northwest is a legacy of the American OSB and plywood industries that grew there. Supply routes into BC and Alberta panel mills run south-to-north, making Canadian forest product manufacturers structurally dependent on a foreign industrial base for a critical input to every tonne of engineered wood product they export.

5.2 Canada's Domestic Resin Manufacturing Reality

Canada operates industrial-scale panel resin manufacturing at Hexion facilities in Edmonton, Alberta (integrated UF/PF and formaldehyde; company statements cite ≈ 900 million lb/yr capacity) and St. Romuald, Quebec (Forest Products Resins segment per Hexion 10-K). These plants serve Canadian and export customers while combined HS 3506+3909 imports remain \$1.76 billion CAD annually, and western Canadian mills still rely heavily on US Pacific Northwest supply chains. Hexion Edmonton imports bulk phenol (consistent with HS 290711 trade data showing \$121M CAD phenol imports, 99% US) while manufacturing PF locally; resorcinol synthesis sits outside its product line.

Structural adhesives for glulam and CLT are supplied from Canadian plants at smaller scale: LRBG Chemical (Longueuil, QC) and Dural (Dorval, QC) hold CCMC listings for PRF and PUR systems, serving the structural-adhesive segment rather than commodity UF for OSB. At the feedstock level, Methanex Corporation, headquartered in Vancouver, operates Canada's only commercial-scale methanol production facility in Medicine Hat, Alberta (Southeast Alberta Chamber of Commerce 2024; Methanex Corporation 2024; UBC Library Research Guides 2024). Formaldehyde is the leading consumer of methanol (ChemAnalyst 2024). Yet Canada's methanol is largely sold into commodity markets rather than used to underpin a domestic resin manufacturing industry at commercial scale.

Canada holds the feedstock advantage and underuses the conversion step relative to its panel output. Methanol and fibre are Canadian; most formulation margin accrues to US and offshore resin suppliers. Figure 5 maps the full chain and the cross-border flows it generates.

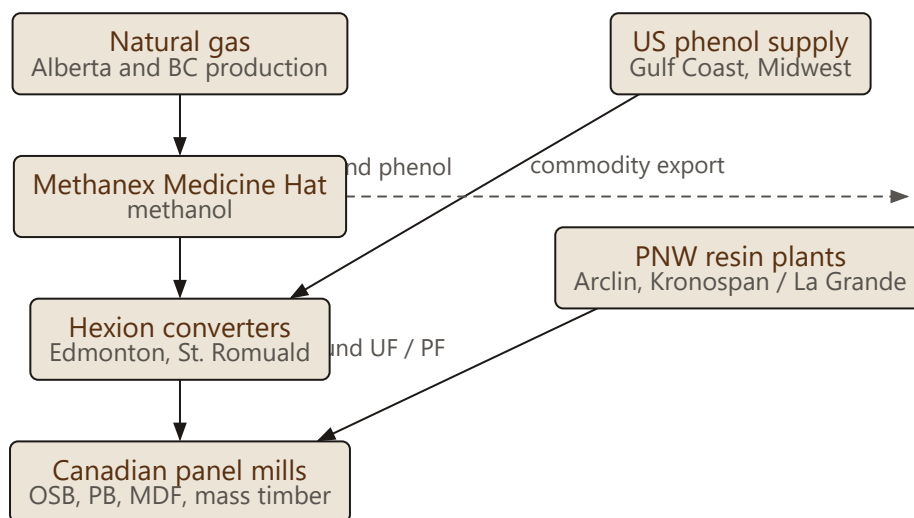


Figure 5: Forest adhesive supply chain serving Canadian panel mills. Canadian assets include natural gas, Methanex methanol, and two Hexion converter plants; US phenol and Pacific Northwest resin capacity fill remaining demand via cross-border import flows quantified in Figure 1 and Table 3. Dashed edge: methanol sold into commodity markets rather than resin conversion at national scale.

At the bio-based alternatives end of the market, there is one notable Canadian innovator: EcoSynthetix (Burlington, Ontario), a publicly traded renewable chemicals company (TSX: ECO) that produces DuraBind™, a no-added formaldehyde, bio-based binder for particleboard, MDF, and OSB (EcoSynthetix Inc. 2023b; 2023c). DuraBind has been adopted by at least one top-15 global wood composites manufacturer (Wood Industry 2023). EcoSynthetix remains a niche player operating at a scale far below the aggregate resin volumes consumed by Canadian panel manufacturers annually.

Table 5: North American forest adhesive and resin manufacturing sites relevant to Canada (illustrative selection).

Site	Operator	Products	Relevance to Canada
Edmonton AB	Hexion	UF/PF, formaldehyde	Only large-scale western converter; imports phenol
St. Romuald QC	Hexion	UF/PF resins	Quebec panel supply; capacity unpublished
Longueuil QC	LRBG	Structural PRF/PUR	CCMC-listed structural segment
Dorval QC	Dural	Structural PRF/PUR	CCMC-listed glulam/CLT adhesives
Burlington ON	EcoSynthetix	Bio-binder	Niche NAUF; below sector volume
Springfield OR	Hexion → Kronospan	UF for boards	PNW supply to BC/AB mills; Dec 2025 acquisition
Portland/Tacoma	Arclin	PF for OSB	Documented Canadian mill customers
La Grande OR	Kronospan	UF captive	Vertical integration; third-party risk

6. Certification Barriers: Drop-In Replacement and Market Entry

Building a resin plant and having that resin accepted by Canadian panel mills are separate problems. Capital, feedstock logistics, and environmental permitting determine whether a facility can be built; CANFER, CSA, and CCMC pathways determine whether mills can switch suppliers without repeating months of laboratory and third-party certification work.

6.1 Two Approval Layers

Layer 1: Resin manufacturing (plant siting). A converter-scale UF/PF resin facility handling formaldehyde, methanol, and phenol is classified under NAICS 325210 (resin and synthetic rubber manufacturing). In British Columbia, new chemical manufacturing facilities that produce Transportation of Dangerous Goods–classified materials at ≥5,000 tonnes/year, or any chemical facility at ≥100,000 tonnes/year, are reviewable under the **Reviewable Projects Regulation** (Province of British Columbia 2019). A Prince Rupert converter in the 50,000–200,000 tonne/year range would likely require provincial environmental assessment, air-emissions authorization, and municipal industrial zoning (M3) compliance before production begins. This is a multi-year permitting sequence distinct from product approval.

Layer 2: Resin use in certified wood products (mill acceptance). Panel and structural-timber manufacturers may not treat a new resin source as interchangeable until their existing product certifications cover it. The approval path depends on product category:

Table 6: Canadian approval pathways for forest adhesive resins by end use.

End use	Governing framework	What is certified	Typical timeline
OSB / waferboard	CANFER + CSA O437.2 / O325 (Canadian Wood Council 2019)	Panel product type + binder performance vs PF benchmark	Months (mill QC + TPC)
Particleboard / MDF	CANFER (SOR/2021-148) (Justice Laws Website 2021)	Product type by resin type and emission profile	Primary chamber test + ongoing QC
Glulam / LVL / CLT	CSA O112.9/O112.10 + CCMC evaluation (CSA Group 2021; National Research Council Canada 2023)	Specific adhesive formulation at specific plant	6–18 months; \$20k+ CCMC fees

6.2 CANFER and the Product-Type Lock-In

Since January 2023, composite wood panels and laminated products sold in Canada must meet formaldehyde emission limits and be certified by a third-party certifier (TPC) under the **Formaldehyde Emissions from Composite Wood Products Regulations** (CANFER) (Justice Laws Website 2021; Health Canada 2023a). CANFER defines a **product type** as panels made by the same manufacturer with the **same type of resin** that differ based on composition and formaldehyde emissions (Justice Laws Website 2021).

This definition places the lock-in at the mill rather than at the resin plant:

1. **Primary testing** requires ASTM E1333 large-chamber testing (or ASTM D6007 with demonstrated equivalence) verified by a TPC (Environment and Climate Change Canada 2023; Justice Laws Website 2021).
2. **Ongoing quality control** may use faster correlated methods once equivalence is established (Environment and Climate Change Canada 2023).
3. **Any change** in manufacturing process or in the amount or formulation of resin that might affect formaldehyde emissions triggers renewed primary and QC testing (Justice Laws Website 2021).
4. **A change in resin type** removes eligibility for reduced-frequency QC testing until the product type re-qualifies under the no-added-formaldehyde (NAUF) or ultra-low-emitting-formaldehyde (ULEF) reduced-frequency provisions (Justice Laws Website 2021).

CANFER accepts TSCA Title VI certification as equivalent (Canada Gazette 2019), which creates a practical path dependency: mills already certified under the US framework can import US-made panels or continue US-supplier resin relationships with minimal incremental Canadian testing, while a new Canadian resin supplier must be integrated into each mill's CANFER product-type documentation, including resin supplier name and address (Justice Laws Website 2021; Health Canada 2023b).

For commodity panels, switching to a new domestic supplier of UF or PF resin is therefore a certification event rather than a simple procurement change. Even where the chemistry is nominally identical, CANFER treats formulation and process changes conservatively. Mills must notify their TPC, repeat emission testing, and may lose reduced-frequency QC benefits if the resin type changes. Canada publishes no resin interchangeability or equivalence-class mechanism that would allow a mill to substitute suppliers based on a shared performance fingerprint alone.

6.3 Structural Adhesives: CSA O112 and CCMC Plant-Specific Evaluation

Structural adhesives for glulam, LVL, and exterior-exposed engineered wood follow a stricter path. CSA O112.9 and O112.10 specify bond durability, creep, and environmental exposure tests for structural wood adhesives (CSA Group 2021). Compliance is demonstrated through CCMC evaluation listings that name the specific adhesive product, the manufacturer, and often the production plant (National Research Council Canada 2024; 2023).

CCMC's Product Assessment Service charges a non-refundable \$3,200 application fee; initial qualification fees range from \$3,500 to \$13,200+ for CCMC services, with third-party testing costs additional; estimated timelines run from 2–6 months for standardized products with existing third-party certification to 8–18 months for highly innovative products (National Research Council Canada 2023). Annual surveillance fees apply.

A CCMC listing for phenol-formaldehyde resin supplied to LVL manufacturers (CCMC 14512-L, issued 2024) illustrates the model: the evaluation covers a named PF resin for use in LVL/PSL at identified production locations (National Research Council Canada 2024). A new Canadian resin plant producing the same chemistry would require its own CCMC assessment (or third-party certification accepted by CCMC) before structural specifiers and AHJs would treat it as code-compliant. Each entrant repeats the full evaluation.

6.4 OSB Binder Qualification (CSA O437.2)

For OSB and waferboard, CSA O437.0 permits any binder that meets CSA O437.2 performance criteria, which compare candidate binders against phenol-formaldehyde benchmarks for moisture, heat, fungal, and creep resistance (CSA Group 1993). CCMC technical guides (e.g., for pMDI binders) require laboratory panel manufacture and physical testing before the binder may be referenced in CSA O325-certified mill production (National Research Council Canada 2015; PFS-TECO 2012).

Binder approval is therefore a laboratory qualification exercise followed by mill-level panel certification through an accredited certification body (e.g., TECO). A new PF or pMDI supply starts its own qualification rather than inheriting an incumbent supplier's CCMC evaluation.

6.5 Supplier Substitution in Practice

Three scenarios clarify the barrier:

- **Scenario A: Same resin type, new Canadian supplier, equivalent formulation.** CANFER requires mill-TPC re-testing when formulation or process might affect emissions; reduced-frequency QC may be suspended pending re-qualification. Commercial risk and testing cost fall on the panel mill rather than the resin start-up.
- **Scenario B: New chemistry (e.g., bio-based NAUF binder).** CANFER sections 10 and 11 provide reduced-frequency testing pathways, but only after 3–6 months of intensive primary and QC testing (13–26 verified tests) (Justice Laws Website 2021). EcoSynthetix-style binders face a long qualification runway before mills achieve economic testing frequency.
- **Scenario C: Importing US-certified resin or panels.** TSCA Title VI cross-acceptance allows CANFER compliance via existing US certification (Canada Gazette 2019), reinforcing incumbent US supply chains.

Incumbent US suppliers benefit from accumulated certifications, established TPC relationships, and TSCA mutual recognition. A greenfield Canadian resin plant must simultaneously finance construction, secure mill offtake, and sponsor per-mill or per-product re-qualification, a barrier distinct from capital cost alone. Figure 6 traces the three qualification routes and shows where the asymmetry enters.

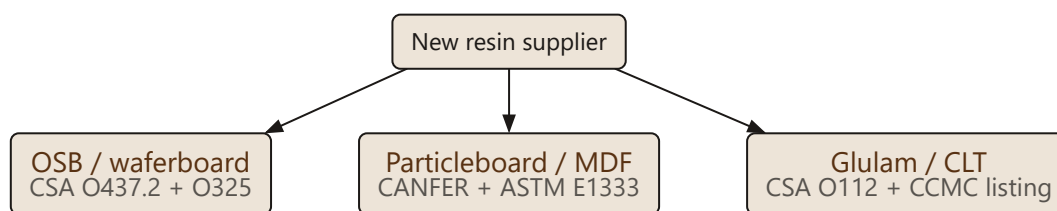


Figure 6: Certification pathways gating mill adoption of a new resin supplier, by product category. Each route imposes time and cost on the adopting mill; TSCA Title VI cross-acceptance admits US-certified supply with minimal incremental Canadian testing, the structural source of certification lock-in. Frameworks: CANFER (SOR/2021-148), CSA O437.2/O325, CSA O112.9/.10, CCMC PAS.

6.6 Toward a Performance-Based QAQC Model

Canada already distinguishes primary certification testing (ASTM E1333 chamber) from routine mill QC (correlated methods such as ASTM D6007) (Environment and Climate Change Canada 2023). The missing piece is a supplier-substitution protocol: a resin-equivalence class system under which a TPC could approve a new supplier when (i) resin chemistry class and solids content fall within a declared envelope, (ii) mill QC correlation to chamber results remains within validated limits, and (iii) a reference emission fingerprint is matched.

Policy options aligned with supply-chain resilience objectives:

1. **NRCan/Health Canada joint guidance** on resin supplier substitution under CANFER, modelled on change-control protocols in pharmaceutical GMP but scoped to emission performance rather than full product-type re-registration.
2. **CCMC fast-track listing** for resin formulations already certified under TSCA Title VI or EU harmonized standards, limiting redundant CSA O112 testing where test data are transferable.
3. **National reference laboratory** for OSB/PB/MDF binder qualification (CSA O437.2) to reduce per-mill duplication when multiple mills adopt the same domestic resin.
4. **IFIT eligibility** for mill re-qualification costs when switching to Canadian resin suppliers, treating certification expense as supply-chain diversification infrastructure.

Without such mechanisms, building domestic resin capacity addresses only half the dependency: mills may remain commercially locked to US-certified supply even when a Canadian alternative exists.

7. Value Capture in Engineered Wood

7.1 Where the Added Value Goes

Canada's forest policy discourse, from federal NRCan programs to BC's provincial strategy, is replete with calls to move "up the value chain," producing more engineered wood products, mass timber, and prefabricated housing components rather than raw lumber and chips (Government of British Columbia 2023a; Business Examiner 2024). The BC government's **Clean and Competitive: Blueprint for BC's Industrial Future** explicitly states that "advanced timber manufacturing creates more jobs for people using less fibre than the traditional forest industry" (Business Examiner 2024).

This is broadly correct. An OSB panel, a glulam beam, or a CLT panel commands significantly higher margins than the raw strands, veneer, or timber they are made from. The value-added manufacturing sector in BC has seen measurable progress (Council of Forest Industries 2025).

The central problem is that the value added to wood by making it into engineered panels is substantially captured in the resin cost, and that resin is largely American. When a BC panel mill converts aspen strands into OSB, roughly 2–6% of what it sends to market is phenol-formaldehyde resin formulated in Oregon or another US state. When a Quebec particleboard plant makes furniture substrate, 9% by weight of that panel is UF resin, most likely sourced from a US production facility.

Every tonne of value-added wood product Canada exports therefore contains a measurable embedded value transfer to the US chemical industry. The Canadian forest sector generates the wood, provides the energy, builds and operates the press lines, pays the labour, and then exports the finished goods with a built-in subsidy to the US specialty chemicals sector.

7.2 The Competitiveness Dimension

The forest sector generates \$87 billion annually and ranks among Canada's top five export industries (Forest Products Association of Canada 2024b). Structural panels, engineered wood, and mass timber are critical to Canada's ambition to solve its housing affordability crisis (Forest Products Association of Canada 2024a). But the competitiveness of Canadian manufacturers in value-added wood products is directly sensitive to the cost and availability of adhesive resins.

Industry analyses of new EWP facility feasibility consistently flag adhesive costs as a key variable, Ontario's engineered wood opportunity study specifically cited adhesive cost sensitivity as one of the key parameters in its financial modelling of new OSB and LSL plants (Ontario Ministry of Natural Resources 2024). The same study confirmed that "all projects can absorb significant shifts in adhesive costs," but noted that adhesive cost volatility has been pronounced (Ontario Ministry of Natural Resources 2024).

That volatility originates outside Canadian control. When US Gulf Coast formaldehyde plants face storms, maintenance shutdowns, or feedstock disruptions, or when US producers make strategic decisions about captive supply, as Kronospan did in acquiring the Hexion La Grande plant (Composite Panel Association 2025; Kronospan 2025), Canadian mills face supply uncertainty and price spikes with few domestic alternatives to fall back on.

Canada's forest products exports face substantial trade pressure: US tariffs have changed the tariff environment, with effective tariff burdens on Canadian softwood lumber approaching 40% (KoalaGains 2025). Canadian forest sector associations have called for urgent supply chain resilience (Forest Products Association of Canada 2024c; 2025). Yet adhesive import dependence, a quieter but equally structural vulnerability, receives almost no policy attention.

7.3 Adhesive Demand from Mass Timber

The exposure grows with each new mass timber plant. Mass timber, cross-laminated timber (CLT), glulam, laminated veneer lumber (LVL), is at the centre of both Canada's housing strategy and its carbon reduction agenda (Forest Products Association of Canada 2024a; The Transition Accelerator 2024). Almost 700 mass timber buildings have been completed in Canada (Forest Products Association of Canada 2024a). The federal government's mass timber roadmap describes the sector as ready to scale, with strong domestic and international demand (The Transition Accelerator 2024).

Mass timber uses different, higher-performance adhesives than commodity panels. CLT is bonded with polyurethane (PUR), emulsion polymer isocyanate (EPI), or melamine-formaldehyde (MF) adhesives (Mass Timber Institute, University of Toronto 2024). Glulam uses PRF or PUR systems (CORRIM 2018b). These are specialty formulations available from a narrower supplier base than commodity UF or PF. Canada is also importing finished mass timber products from Europe alongside its domestic production (Forest Products Association of Canada 2024a).

A country that struggles to supply the adhesive chemistry for commodity panel manufacturing starts even further back in developing the specialty adhesive supply chains that mass timber growth demands.

Demand for engineered wood adhesives is growing with mass timber and housing prefabrication (Forest Products Association of Canada 2024a). Canadian producers will participate in that growth primarily as customers of imported resin systems unless domestic converter capacity expands.

8. The La Grande Transaction

The Hexion La Grande facility has long been recognized as a key resin manufacturing node for the North American panel industry, manufacturing UF resins specifically for board production (Chemical Week 2025; EUWID Wood Products & Panels 2025). Its December 2025 acquisition by Kronospan, a European panel giant that operates US facilities in Alabama, Louisiana, Oregon, and Pennsylvania (Kronospan 2025), is a notable market-structure change that illustrates both the fragility and the asymmetry of Canada's position.

Kronospan, as a vertically integrated panel manufacturer, now owns the resin plant that supplies its own Oregon particleboard plant, and potentially other customers (Composite Panel Association 2025; Kronospan 2025). The strategic logic is clear: securing captive supply chains by bringing chemistry in-house. This is precisely the kind of vertical integration that strengthens Kronospan's position in the North American market while weakening the supply security available to Canadian mills that previously relied on third-party resin availability from that facility.

The direction runs one way. US-based panel producers are integrating upstream into chemistry; Canadian-based panel producers lack comparable chemistry assets to integrate around.

9. The Case for Prince Rupert

9.1 Port, Rail, and Site Attributes

Any serious discussion of where to locate a Canadian forest adhesive resin manufacturing complex must grapple with supply chain geography. We present Prince Rupert as a site screening case, geographic and logistics suitability only; financial feasibility and NPV analysis remain future work. The base chemicals for formaldehyde-based resins, methanol, urea, phenol, and melamine, are sourced globally, with China being the world's largest producer of methanol and a major supplier of key intermediates (Market Growth Reports 2024; Chemical Market Analytics 2024).

Canada has one existing methanol producer (Methanex, Medicine Hat) (Southeast Alberta Chamber of Commerce 2024), but Medicine Hat is positioned to serve Prairie and Interior markets. For a resin plant intended

to supply BC's major OSB, MDF, and mass timber complex, as well as exploit Asia-facing trade flows, the logical geography converges on Prince Rupert.

Prince Rupert is situated at the head of the CN mainline on the North Coast of British Columbia, adjacent to the fastest-growing port in North America and the third-largest port in Canada (City of Prince Rupert 2024). The port is three days closer to Asian markets than Vancouver (City of Prince Rupert 2024). CN Rail's mainline bisects the Watson Island site, meaning bulk chemicals arriving by ship from Asia can transfer directly to rail, and finished resin can be dispatched east to serve Interior BC mills, or exported west to Asian customers (City of Prince Rupert 2024).

Watson Island is a remediated brownfield industrial site. The former Skeena Cellulose pulp mill site (operating 1955–2001) has been redeveloped, and the City of Prince Rupert is actively marketing 75 free acres of the industrial park plus 2,200 acres of adjacent greenfield at Lot 444 (City of Prince Rupert 2024; Business in Vancouver 2024; North Coast Review 2024). The Watson Island site features:

1. CN mainline with spur options, direct rail to Interior BC and national markets (City of Prince Rupert 2024)
2. Deep-water marine access; precursor chemicals arrive by sea from Asia (City of Prince Rupert 2024)
3. Private hydro substation with BC Hydro agreements for future capacity (Business in Vancouver 2024)
4. 3 m³/second water supply, industrial-scale process water (City of Prince Rupert 2024)
5. Pre-zoned M3 marine industrial, suitable for chemical manufacturing (City of Prince Rupert 2024)
6. Established relationships with Ts'msyen nations and First Nations Forestry frameworks (City of Prince Rupert 2024)
7. City-owned and development-ready through Prince Rupert Legacy Inc. (North Coast Review 2024)

The Canada Infrastructure Bank has already committed \$150 million to the Port of Prince Rupert for the CANXPORT export logistics hub (Canada Infrastructure Bank 2024). The Province of BC has invested up to \$11.5 million in the Port of Prince Rupert's Integrated Marketplace initiative (Government of British Columbia 2023b).

Prince Rupert offers logistics attributes relevant to a resin manufacturing facility that would: (1) import precursor chemicals (methanol, urea, phenol, melamine) from Asia by sea; (2) manufacture UF, MUF, PF, and/or pMDI-compatible resins at commercial scale; (3) distribute finished resins east via CN rail to Interior BC, Alberta, and Saskatchewan panel mills; and (4) potentially export specialty bio-based or low-emission resins to Asian markets.

9.2 The Chemistry Logic

The formaldehyde-based resin manufacturing process is well-understood and involves a straightforward two-step process: methanol conversion to formaldehyde, followed by condensation with urea, phenol, or melamine in batch reactors (CORRIM 2018b). This is mature industrial chemistry that dozens of plants in the US, Europe, and Asia have been running for decades, at capital intensities well below petrochemical synthesis.

The facility would serve as a converter/formulator, much like the La Grande plant does: importing bulk commodity chemicals and converting them into the specific panel-grade resin formulations demanded by wood composites manufacturers. The capital intensity is moderate relative to, say, a petrochemical cracker or an LNG terminal.

A Prince Rupert resin plant could shorten supply routes relative to Pacific Northwest US sources. Rather than Canadian mills trucking or railing resin 1,200–1,500 km from Oregon or Washington, they could receive

supply from a Canadian facility at the western end of the CN mainline corridor, subject to Canadian commercial and regulatory frameworks.

10. Policy Context

10.1 Tariffs and the Investment Window

The 2025 US tariff regime has changed the calculus for Canadian forest sector investment. US tariffs on Canadian products, approaching 25% blanket tariffs on many goods, are forcing Canada to confront the degree of its supply chain exposure (Miller Thomson 2025; University of Guelph 2025). Forestry products, at \$38 billion in US-bound exports and 81% dependence on the US market (Miller Thomson 2025), are among the most exposed sectors.

These tariffs open a window for domestic industrial policy that was closed while cross-border supply ran smoothly. When Canada's forest sector must diversify markets and supply chains regardless, the argument for domestic chemical manufacturing capacity becomes far easier to make as urgent supply chain resilience.

The forest sector itself has formally called on the federal government to “expand domestic wood use,” “strengthen trade resilience,” and “cut red tape and regulatory barriers” (Forest Products Association of Canada 2025). The regulatory framework exists to support such investment: NRCan's Investments in Forest Industry Transformation (IFIT) program explicitly funds capital investments for “expanding domestic housing production capacity to resolve supply chain bottlenecks” and “reducing costs and improving resilience and innovation in housing construction supply chains” (Natural Resources Canada 2024d). A domestic resin plant would squarely meet these objectives.

10.2 Regulatory Drivers for Low-Emission Resins

Canada implemented the **Formaldehyde Emissions from Composite Wood Products Regulations** (SOR/2021-148) in 2021, entering into force January 2023 (SGS 2021). The regulations mirror California's CARB Phase 2 limits and require manufacturers and importers of composite wood products to meet strict formaldehyde emission thresholds (Health Canada 2023a; Canada Gazette 2019).

Critically, the regulations require manufacturers to disclose the names and addresses of resin suppliers (Justice Laws Website 2021). This creates a documentary record of the degree to which Canadian composite wood manufacturing is dependent on foreign resin supply, a record that, if properly analyzed, would likely reveal the extent of the Oregon/Pacific Northwest dependency.

These same regulations create a commercial incentive to adopt lower-emission resin systems: UF resins with scavengers, MUF formulations, pMDI systems, or bio-based alternatives like DuraBind (EcoSynthetix Inc. 2023b; Floor Covering Weekly 2024). A Canadian resin manufacturer at Prince Rupert would be ideally positioned to manufacture next-generation low-emission resins tailored to Canadian regulatory requirements, rather than adapting US formulations designed for CARB compliance.

10.3 The Bio-Based Dimension

Canada holds real innovation assets in this space. EcoSynthetix has developed a commercially proven bio-based no-added formaldehyde binder system achieving competitive performance and economics relative to incumbent UF/MUF systems (EcoSynthetix Inc. 2023b; 2023a). The cost profile (approximately \$1.53/unit vs \$2.26 for UF and \$5.09 for pMDI) positions DuraBind as genuinely competitive (EcoSynthetix Inc. 2023a).

Meanwhile, Hexion's Edmonton R&D facility has been focused on developing next-generation resin chemistry that uses lignin, a forestry byproduct, to replace petrochemical-based ingredients in OSB resins (Hexion Inc. 2024a; 2024b). This work has direct relevance to British Columbia, which produces enormous volumes of kraft pulping lignin that is currently burned for energy rather than valorized as a resin feedstock.

A Prince Rupert resin complex could credibly incorporate a bio-based component by sourcing lignin from BC pulp mills as a partial replacement for phenol in PF resins, a substitution strategy already validated at laboratory and pilot scale (FPInnovations 2020b).

11. The Scale of Foregone Value Capture

Canada's wood panel manufacturing sector generates revenue of \$11.8 billion annually (IBISWorld 2025). The forest sector overall contributes \$27 billion to GDP and exports \$36.2 billion in forest products (Natural Resources Canada 2024b). Canada's structural panel production, OSB and softwood plywood combined, reached approximately 9.9 billion square feet in 2024 (annualized from APA H1 data) (Fastmarkets 2024a).

Trade data confirm that \$1.28 billion CAD of the \$1.76 billion in annual adhesive and bulk-resin imports originated in the United States in 2023 (Statistics Canada 2024), value that flows primarily to US specialty chemical producers through Pacific Northwest and Gulf Coast supply chains. The bottom-up model across OSB, plywood, particleboard, and fibreboard implies \$1.16 billion in annual resin demand (CORRIM 2018b; Food and Agriculture Organization of the United Nations 2024), bracketing the forest sector's adhesive cost between \$1.2 billion and \$1.8 billion depending on scope and product mix.

Meanwhile, Canada's wood panel manufacturing industry revenues have grown at a CAGR of only 1.5% (IBISWorld 2025). The low growth rate is partly a reflection of commodity pricing exposure, but it also reflects the reality that the chemical margin embedded in each panel is captured upstream by someone else.

A purpose-built resin manufacturing facility at Prince Rupert, sized to supply BC and Alberta's panel mills, would:

1. Displace imports of UF, MUF, and PF resins currently sourced from Pacific Northwest US plants
2. Reduce transport costs for resin delivery to Interior BC and Alberta mills (Prince Rupert to Prince George by rail is ~760 km vs. La Grande, OR to Prince George ~1,100 km by road/rail)
3. Create a domestic pricing benchmark that prevents price-taking behaviour in resin procurement
4. Use BC and Alberta natural gas via local methanol production (Methanex Medicine Hat as existing supply)
5. Enable formulation innovation tailored to Canadian wood species and Canadian regulatory standards
6. Anchor an industrial chemistry cluster at Watson Island expandable into specialty mass-timber adhesives (PUR, EPI, MF systems)

12. Remaining Data Gaps

Primary 2023 trade totals, US shares, a 2014–2024 time series, disaggregated HS 390940/390950/290711/290721 flows, particleboard and fibreboard volumes, phenol supply-chain history, resorcinol routing, and a North American plant map are documented in the supplementary workbook (Table 2, Table 3, Table 5). The following gaps remain for peer-reviewed submission:

1. **Full-year APA yearbook volumes:** would replace the H1-annualized structural panel figures and refresh particleboard/fibreboard beyond FAO 2022
2. **CBSA importer-level data:** Top importers cross-referenced with Pacific Northwest resin plant locations
3. **ISED NAICS 32552:** Domestic resin production statistics to complement import totals
4. **Kronospan La Grande capacity:** Published plant throughput and historical Canadian customer share
5. **Watson Island tenants:** Confirm RFP status with City of Prince Rupert Economic Development (City of Prince Rupert 2024)

13. What the Findings Mean by Audience

The trade and certification findings restate differently for researchers, forest-sector executives, and trade policymakers. No new numbers are introduced here.

13.1 For the Researcher

The dependency claim now rests on two independent estimation methods that bracket each other (Figure 3), and the residual uncertainty is concentrated in scope rather than magnitude. The highest-value refinements are: (i) mass-timber adhesive volumes (glulam, CLT, LVL), which would raise the lower bound K in (2) further and narrow the bracket; (ii) CBSA importer-level microdata, which would convert the inference that HS 3506/3909 flows serve panel mills into a direct observation; and (iii) the CANFER disclosure requirement, under which every certified mill must already report its resin supplier's name and address (Justice Laws Website 2021), a national census of resin sourcing sitting in TPC files that awaits only aggregation authority.

13.2 For the Forest Sector Executive

The operational risk is concentrated US routing combined with certification lock-in. Three points follow from (3) and Figure 3:

- **Import reliance exceeds \$1.7B/yr and is rising.** Combined HS 3506+3909 imports grew 72% from 2014 to 2024 (Figure 1). A mill procurement officer budgeting resin cost should work with the \$1.2–1.8B forest-sector bracket rather than the \$792M structural-panel component alone.
- **Western mills face Pacific Northwest single routing.** HS 3506 prepared-adhesives share has remained 81–86% US throughout the decade (Figure 2). Kronospan's December 2025 acquisition of Hexion La Grande converts a third-party supplier into a captive competitor's asset, tightening supply for independent BC and Alberta mills.

- **Approval cost blocks drop-in substitution.** The chemistry transfers readily; the certification burden stays. Figure 6 documents the two-layer gate (CANFER product-type lock-in, CSA O112/CCMC plant-specific evaluation). A domestic resin entrant must secure per-mill re-qualification before displacing Oregon supply; procurement contracts should treat certification-contingent offtake as a pre-construction condition.

13.3 For the Trade Policymaker

The policy problem decomposes into two halves that must move together. Building capacity without certification reform leaves mills commercially locked to US-certified supply even when a Canadian alternative exists (Section 6.6); certification reform without capacity simply eases imports. The instruments are already identified in Section 6.7 and the Recommendations: a CANFER supplier-substitution protocol (regulatory guidance, near-zero fiscal cost), CCMC fast-tracking for TSCA/EU-certified formulations, IFIT eligibility for both plant capital and mill re-qualification expenses, and a national reference laboratory for binder qualification.

The asymmetry to correct is precise: the current framework charges a Canadian entrant months and per-mill fees for market access that US incumbents receive through TSCA cross-acceptance at nearly zero marginal cost (Figure 6). Canada's own regulations subsidize the import dependence its industrial policy seeks to reduce. Tariff and trade-diversification strategies aimed at lumber alone (HS 44) leave untouched the northbound resin routing documented in Figure 4. A coherent forest-products industrial policy must treat HS 3506/3909 import reliance with the same urgency as softwood lumber duties.

14. Recommendations

For Federal and Provincial Policy Makers:

1. Commission a formal supply chain vulnerability assessment of the Canadian forest products sector's adhesive resin dependency, disaggregated by resin type, panel product, and province (NRCan/ISED joint initiative with FPAC engagement).
2. Include domestic adhesive resin capacity as an eligible investment category under the NRCan IFIT program renewal (2026), alongside mass timber and prefabricated housing (Natural Resources Canada 2024d).
3. Commission an independent site-selection study for domestic converter-scale resin capacity, including Prince Rupert (Watson Island) as one Pacific routing option (City of Prince Rupert 2024; Canada Infrastructure Bank 2024).
4. Direct Statistics Canada to publish a focused analysis of HS 3506 and 3909 import flows broken out by importing industry sector.
5. Task Health Canada and the NRC (CCMC) with developing a CANFER resin-supplier substitution protocol so mills can adopt equivalent domestic resins without repeating full primary chamber certification for unchanged emission profiles.
6. Clarify BC environmental assessment thresholds for converter-scale resin plants (NAICS 325210) early in site-selection studies (Province of British Columbia 2019).

For Industry:

1. FPAC and COFI should jointly commission an adhesive supply chain risk study modelling exposure of member mills to US resin supply disruptions, including implications of the Kronospan/Hexion La Grande transaction (Composite Panel Association 2025).
2. Major BC and Alberta panel producers should explore a joint venture or co-operative procurement model to underwrite the offtake agreement that would make a Prince Rupert resin plant financially viable.
3. EcoSynthetix and FPIInnovations should be engaged to develop a roadmap for integrating lignin-derived phenol substitutes into a BC-based resin manufacturing operation (FPIInnovations 2020b; EcoSynthetix Inc. 2023b).

15. Conclusion

Canada is one of the world's great forest nations. Its panel and timber industries are sophisticated, and its policy direction is unambiguous: more value added, more engineered products, more mass timber, more domestic housing output.

Part of that added value leaves the country with each shipment, because the critical binding agent is manufactured in Oregon, distributed from warehouses in Washington and Oregon, and priced according to US market dynamics. Every OSB panel, every particleboard sheet, every MDF board, every glulam beam that leaves a Canadian mill contains a small but measurable contribution to the US chemical industry.

Structural import dependence, concentrated among a handful of US suppliers and reinforced by certification pathways, limits domestic value capture in wood composites. In the context of 2025–2026 tariff disruptions, the Kronospan/Hexion La Grande transaction, rising mass timber demand, and available industrial land at Prince Rupert, the timing of policy and investment choices merits attention.

The glue that holds Canadian forest products together should be made in Canada.

16. Acknowledgment

The author is principal of Newcopia (2809826 Alberta Inc.). This analysis is independent research. The author acknowledges use of publicly available materials from the City of Prince Rupert regarding Watson Island industrial development; no financial relationship with the City or port authority is declared.

17. Data Availability

The complete trade dataset, the 2014–2024 CIMT import series, partner-country breakdowns, precursor chemical flows (phenol and resorcinol), the bottom-up resin demand model, and the North American plant map, accompanies this paper as a supplementary Excel workbook cited in the Methods section. Updates on request from john@newcopia.ca. Open-access PDF: <https://doi.org/10.5281/zenodo.21500239>.

17. References

- AdvanSix Inc. 2024. "Frankford, PA Manufacturing Site.". <https://www.advansix.com/about/manufacturing-sites/frankford-pa/>.
- Arclin. 2026a. "Arclin Completes Acquisition of the Willamette Valley Company".
- Arclin. 2026b. "Arclin Newsroom".
- Arclin Inc. 2019. "Arclin 5440 Technical Data Sheet".
- Borregaard. 2024. "How to Reduce Manufacturing Costs in Wood-Based Panel".
- Business Examiner. 2024. "Honest Commentary on British Columbia's Efforts to Grow Value-Added Wood Products Manufacturing".
- Business in Vancouver. 2024. "Prince Rupert Seeks New Life for Former Pulp Mill Site".
- Canada Gazette. 2019. "Formaldehyde Emissions from Composite Wood Products Regulations".
- Canada Infrastructure Bank. 2024. "CIB Investment to Improve Flow of Canadian Exports Through Port of Prince Rupert".
- Canadian Wood Council. 2019. "Oriented Strand Board (OSB) Grades".
- ChemAnalyst. 2024. "Canada Methanol Market Size, Share, Growth & Forecast".
- Chemcess. 2024. "Phenol: Production, Reactions and Uses". <https://chemcess.com/phenol/>.
- Chemical Market Analytics. 2024. "World Analysis – Formaldehyde & Resins".
- Chemical Week. 2025. "Hexion to Sell Oregon Manufacturing Site".
- City of Prince Rupert. 2024. *Watson Island and Lot 444 Information Sheet*. Technical report.
- Composite Panel Association. 2024. "Arclin Member Profile".
- Composite Panel Association. 2025. "Kronospan Purchases Hexion Chemical Plant".
- CORRIM. 2018b. *A Life-Cycle Inventory of Manufacturing Resins Used in the Wood Composites Industry*. Technical report.
- CORRIM. 2018a. *Particleboard: A Life-Cycle Inventory of Manufacturing Panels from Wood Resources*. Technical report.
- Council of Forest Industries. 2025. *Forestry Is a Value-Added Industry*. Technical report.
- Covestro. 2020. "A Life Cycle Assessment of Particle Board: UF Vs. MDI as the Adhesive".
- CSA Group. 1993. "CSA O437.2: Evaluation of Binder Systems for OSB and Waferboard".
- CSA Group. 2021. "CSA O112.9: Evaluation of Adhesives for Structural Wood Products (Exterior Exposure)".
- EcoSynthetix Inc. 2023b. *Durabind Bio-Based Binders: A Healthy & Sustainable Solution*. Technical report.
- EcoSynthetix Inc. 2023a. *Durabind Factsheet: Cost Comparison*. Technical report.
- EcoSynthetix Inc. 2023c. "Ecosynthetix Announces Ongoing Commercial Durabind Volumes with Top 15 Wood Composites Manufacturer".
- Environment and Climate Change Canada. 2023. "Directive Concerning Testing for Formaldehyde Emissions (June 2023)".
- EUWID Wood Products & Panels. 2025. "Hexion Sells Resin Production in La Grande to Kronospan".
- Fastmarkets. 2024a. "North American Panel Production Falls in First Half Compared to 2023".
- Fastmarkets. 2024b. "North American Panel Production Falls in First Half Compared to 2023".
- Floor Covering Weekly. 2024. "Ecosynthetix Takes on Formaldehyde with Eco-Friendly Binder".
- Food and Agriculture Organization of the United Nations. 2024. "Forestry Production and Trade". <https://www.fao.org/faostat/en/>.
- Forest Products Association of Canada. 2024b. *Improving Regulatory Efficiency: A One-Window Solution for Canada's Forest Sector*. Technical report.
- Forest Products Association of Canada. 2024a. *The Mass Timber Roadmap*. Technical report.
- Forest Products Association of Canada. 2024c. "Forest Products Sector Calls for Action on Critical Supply Chain Issues".

- Forest Products Association of Canada. 2025. "Canada's Forest Sector Responds to Threats of U.S. Tariffs on Canadian Wood Products."
- Forisk Consulting. 2024. "Sizing up Canada's Forest Products Industry."
- FPInnovations. 2020a. *Effect of MDI and Powder PF Combination Binder System on OSB*. Technical report.
- FPInnovations. 2020b. *MDF and Particleboard with Ultra Low Formaldehyde Emissions*. Technical report.
- Genome Québec. 2019. *The Forestry Value Chain: Stakeholder Engagement and Strategy*. Technical report.
- Government of British Columbia. 2023b. "Province Partners with Port of Prince Rupert for Cleaner, Greener Trade."
- Government of British Columbia. 2023a. "Value-Added Wood Manufacturing Expanded Through Collaboration."
- Health Canada. 2023a. "Guidance Document on the Formaldehyde Emissions from Composite Wood Products Regulations."
- Health Canada. 2023b. "Third Party Certifiers: Composite Wood Products Containing Formaldehyde."
- Health Canada / Environment Canada. 1997. "Priority Substances List Assessment Report for Phenol." <https://www.canada.ca/en/health-canada/services/environmental-workplace-health/reports-publications/environmental-contaminants/canadian-environmental-protection-act-1999-priority-substances-list-assessment-report-phenol.html>.
- Hexion Inc. 2020. "Hexion's Critical Role in North American Panel Industry."
- Hexion Inc. 2024a. "Hexion Edmonton R&D Facility."
- Hexion Inc. 2024b. "The Next Generation: Hexion R&D Facility."
- IBISWorld. 2025. "Wood Panelling Manufacturing in Canada Industry Analysis, 2025."
- INEOS Phenol. 2018. "INEOS Phenol to Expand Capacity in Mobile, Alabama." <https://www.fuelsandlubes.com/ineos-phenol-expand-capacity-mobile-ala-plant-making-largest-phenol-production-unit-world/>.
- INIS/IAEA. 1990. "The Petrochemical Industry in Quebec: From Yesterday to Today." <https://inis.iaea.org/records/dmw81-b4727>.
- Justice Laws Website. 2021. "Formaldehyde Emissions from Composite Wood Products Regulations (Sor/2021-148)."
- KoalaGains. 2025. "Forest Products Industry Tariff Report: 2025 US Policy."
- Kronospan. 2025. "Kronospan Completes Acquisition of Hexion's Chemical Plant in La Grande, Oregon."
- Li, Y. 2010. "The Use of P-MDI Resin in MDF Manufacture." Master's thesis.
- Market Growth Reports. 2024. "Wood Based Panel Market Size, Share."
- Mass Timber Institute, University of Toronto. 2024. "FAQ – Mass Timber Institute."
- Methanex Corporation. 2024. "Methanex – Vancouver Headquarters."
- Miller Thomson. 2025. "2025 U.S. Tariff Threats: A Legal Roadmap for Canadian Business Resilience."
- National Research Council Canada. 2015. "CCMC 14125-R: MONDUR 541-LIGHT Binder for Osb."
- National Research Council Canada. 2023. "CCMC Product Assessment Service (PAS-001)."
- National Research Council Canada. 2024. "CCMC 14512-L: Phenol-Formaldehyde Resin for Structural Wood Products."
- Natural Resources Canada. 2024a. *10 Key Facts on Canada's Natural Resources – 2024*. Technical report.
- Natural Resources Canada. 2024b. "How Does the Forest Sector Contribute to Canada's Economy?."
- Natural Resources Canada. 2024d. "Investments in Forest Industry Transformation (Ifit)."
- Natural Resources Canada. 2024c. "State of Canada's Forests: Forest Products Trade."
- North Coast Review. 2024. "Mayor Pond Promotes Watson Island as Energy Hub at Vancouver Forum."
- Ontario Ministry of Natural Resources. 2024. *Opportunity Study of Hardwood-Based Engineered Wood Products*. Technical report.
- PatSnap Eureka. 2024. "Phenol Formaldehyde Oriented Strand Board Adhesive."
- PFS-TECO. 2012. "OSB Gradestamp: CSA O325 and CSA O437."
- Province of British Columbia. 2019. "Reviewable Projects Regulation (B.C. Reg. 243/2019)."
- S&P Global / IHS Markit. 2020. "Process Economics Program: Integrated Cumene-Phenol/acetone." https://cdn.ihsmarkit.com/www/pdf/0720/RW2020-10_toc.pdf.
- SGS. 2021. "Canada Regulates Formaldehyde Emissions from Composite Wood Products."

-
- Southeast Alberta Chamber of Commerce. 2024. "Methanex Corporation – Medicine Hat."
- Statistics Canada. 2024. "Canadian International Merchandise Trade (Customs Basis): HS Imports."
- The Transition Accelerator. 2024. *The Mass Timber Roadmap*. Technical report.
- UBC Library Research Guides. 2024. "Methanex Company Profile."
- University of Guelph. 2025. "How the Canada-U.S. Tariff War Will Impact Products and Industries."
- Wood Industry. 2023. "Ecosynthetix Durabind Resin Implemented by Wood Composite Leader."